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RHRP15120

Data Sheet **November 2013**

15 A, 1200 V, Hyperfast Diode

The RHRP15120 is a hyperfast diode with soft recovery characteristics. It has the half recovery time of ultrafast diodes and is silicon nitride passivated ionimplanted epitaxial planar construction. These devices are intended to be used as freewheeling/ clamping diodes and diodes in a variety of switching power supplies and other power switching applications. Their low stored charge and hyperfast soft recovery minimize ringing and electrical noise in many power switching circuits reducing power loss in the switching transistors.

Features

- Hyperfast Recovery t_{rr} = 75 ns (@ I_F = 15 A)
- Max Forward Voltage, V_F = 3.2 V (@ T_C = 25°C)
- 1200 V Reverse Voltage and High Reliability
- Avalanche Energy Rated
- RoHS Compliant

Applications

- Switching Power Supplies
- Power Switching Circuits
- General Purpose

Ordering Information

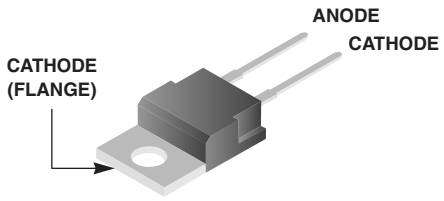
PART NUMBER	PACKAGE	BRAND
RHRP15120	TO-220AC-2L	RHR15120

NOTE: When ordering, use the entire part number.

Symbol

Packaging

JEDEC TO-220AC



Absolute Maximum Ratings T_C = 25°C, Unless Otherwise Specified

		RHRP15120	UNIT
Peak Repetitive Reverse Voltage	V_{RRM}	1200	V
Working Peak Reverse Voltage	V_{RWM}	1200	V
DC Blocking Voltage	V_R	1200	V
Average Rectified Forward Current (T_C = 140°C)	$I_{F(AV)}$	15	A
Repetitive Peak Surge Current (Square Wave, 20 kHz)	I_{FRM}	30	A
Nonrepetitive Peak Surge Current (Halfwave, 1 Phase, 60 Hz)	I_{FSM}	200	A
Maximum Power Dissipation	P_D	100	W
Avalanche Energy (See Figures 10 and 11)	E_{AVL}	20	mJ
Operating and Storage Temperature	T_{STG}, T_J	-65 to 175	°C

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October-2017, Rev. 3

Publication Order Number:
RHRP15120/D

Electrical Specifications $T_C = 25^oC$, Unless Otherwise Specified

SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
V_F	I_F = 15 A	-	-	3.2	V
	I_F = 15 A, $T_C = 150^oC$	-	-	2.6	V
I_R	V_R = 1200 V	-	-	100	μA
	V_R = 1200 V, $T_C = 150^oC$	-	-	500	μA
T_{rr}	I_F = 1 A, dI_F/dt = 100 A/μs	-	-	65	ns
	I_F = 15 A, dI_F/dt = 100 A/μs	-	-	75	ns
t_a	I_F = 15 A, dI_F/dt = 100 A/μs	-	36	-	ns
t_b	I_F = 15 A, dI_F/dt = 100 A/μs	-	28	-	ns
Q_{rr}	I_F = 15 A, dI_F/dt = 100 A/μs	-	150	-	nC
C_J	V_R = 10 V, I_F = 0 A	-	55	-	pF
$R_{\theta JC}$		-	-	1.5	$^oC/W$

DEFINITIONS

V_F = Instantaneous forward voltage (pw = 300 μs, D = 2%).

I_R = Instantaneous reverse current.

t_{rr} = Reverse recovery time (See Figure 9), summation of $t_a + t_b$.

t_a = Time to reach peak reverse current (See Figure 9).

t_b = Time from peak I_{RM} to projected zero crossing of I_{RM} based on a straight line from peak I_{RM} through 25% of I_{RM} (See Figure 9).

Q_{rr} = Reverse recovery charge.

C_J = Junction capacitance.

$R_{\theta JC}$ = Thermal resistance junction to case.

pw = pulse width.

D = duty cycle.

Typical Performance Curves

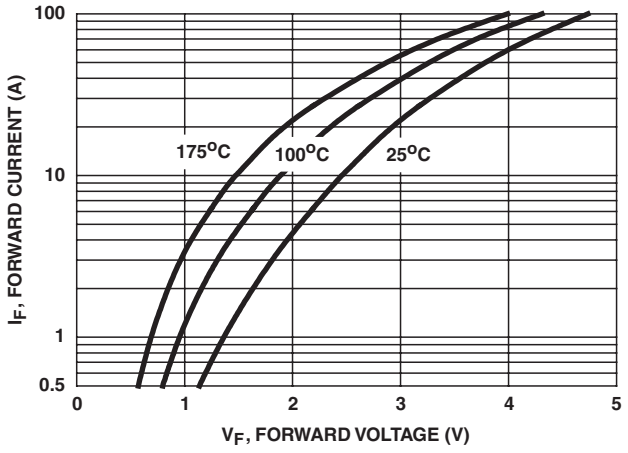

FIGURE 1. FORWARD CURRENT vs FORWARD VOLTAGE

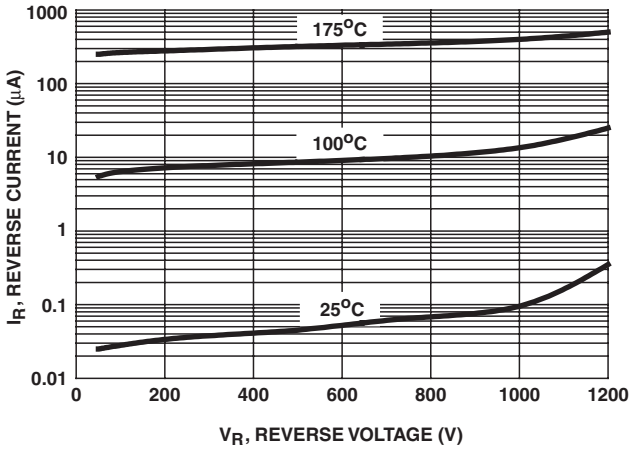

FIGURE 2. REVERSE CURRENT vs REVERSE VOLTAGE

Typical Performance Curves (Continued)

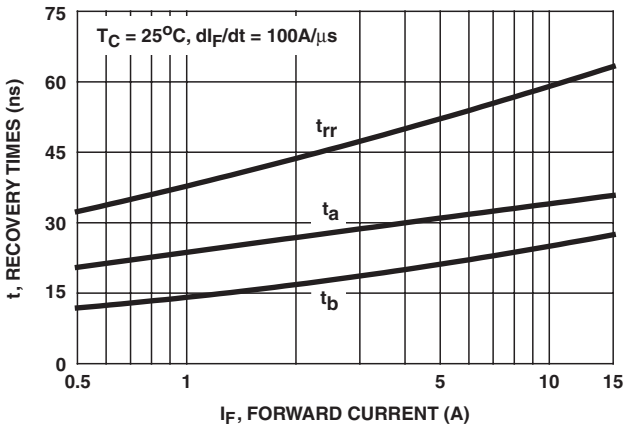

FIGURE 3. t_{rr}, t_a AND t_b CURVES vs FORWARD CURRENT

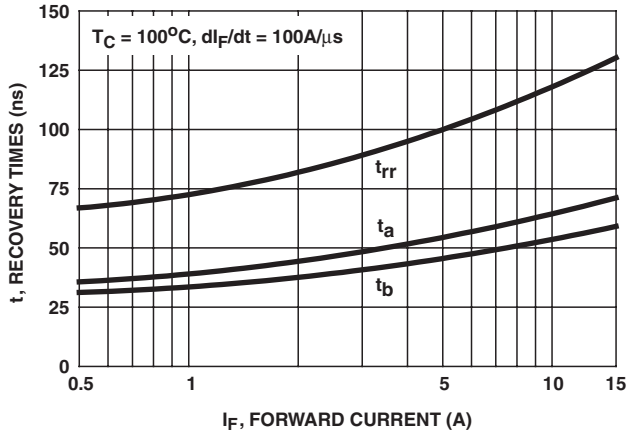

FIGURE 4. t_{rr}, t_a AND t_b CURVES vs FORWARD CURRENT

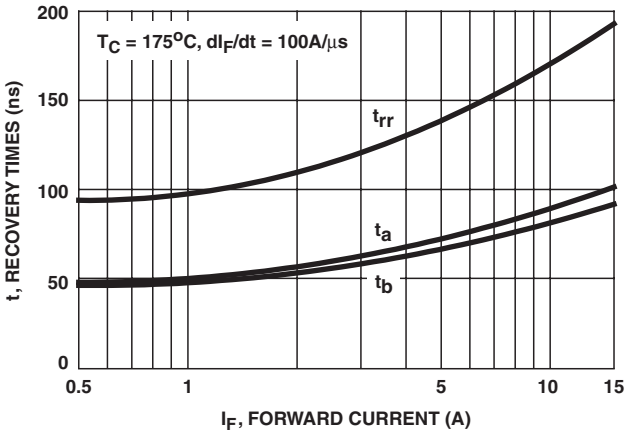

FIGURE 5. t_{rr}, t_a AND t_b CURVES vs FORWARD CURRENT

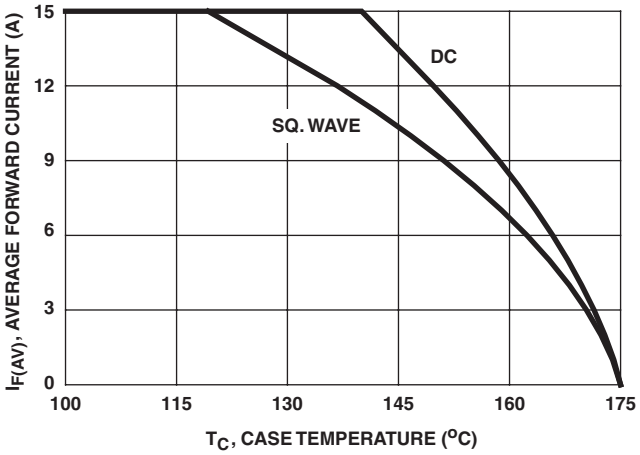

FIGURE 6. CURRENT DERATING CURVE

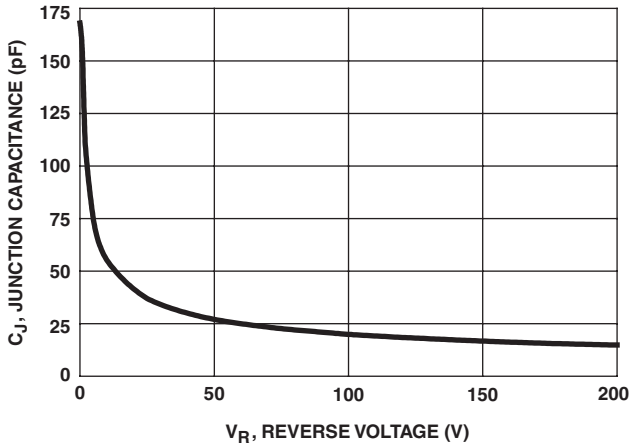

FIGURE 7. JUNCTION CAPACITANCE vs REVERSE VOLTAGE

Test Circuits and Waveforms

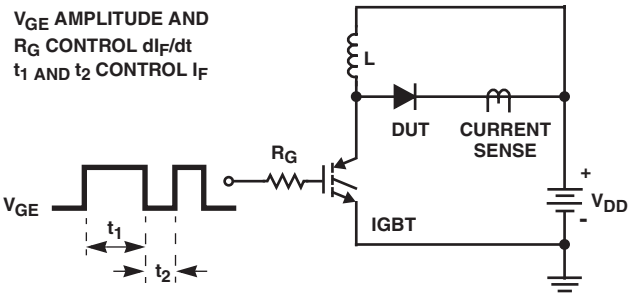

FIGURE 8. t_{rr} TEST CIRCUIT

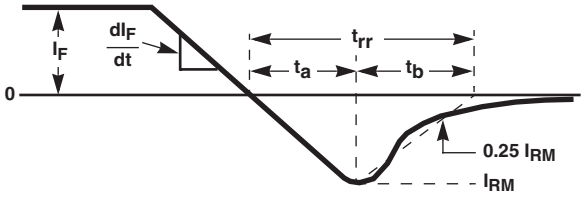

FIGURE 9. t_{rr} WAVEFORMS AND DEFINITIONS

I_{MAX} = 1A
L = 40mH
R < 0.1Ω
$E_{AVL} = 1/2LI^2 [V_{R(AVL)}/(V_{R(AVL)} - V_{DD})]$
Q_1 = IGBT (BV_{CES} > DUT $V_{R(AVL)}$)

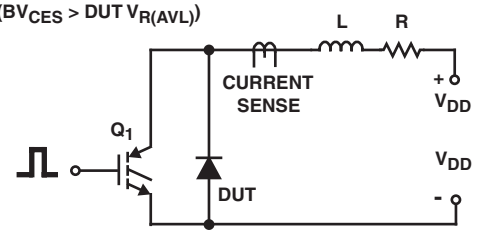

FIGURE 10. AVALANCHE ENERGY TEST CIRCUIT

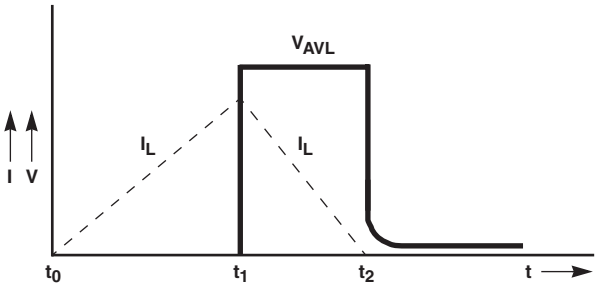

FIGURE 11. AVALANCHE CURRENT AND VOLTAGE WAVEFORMS

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